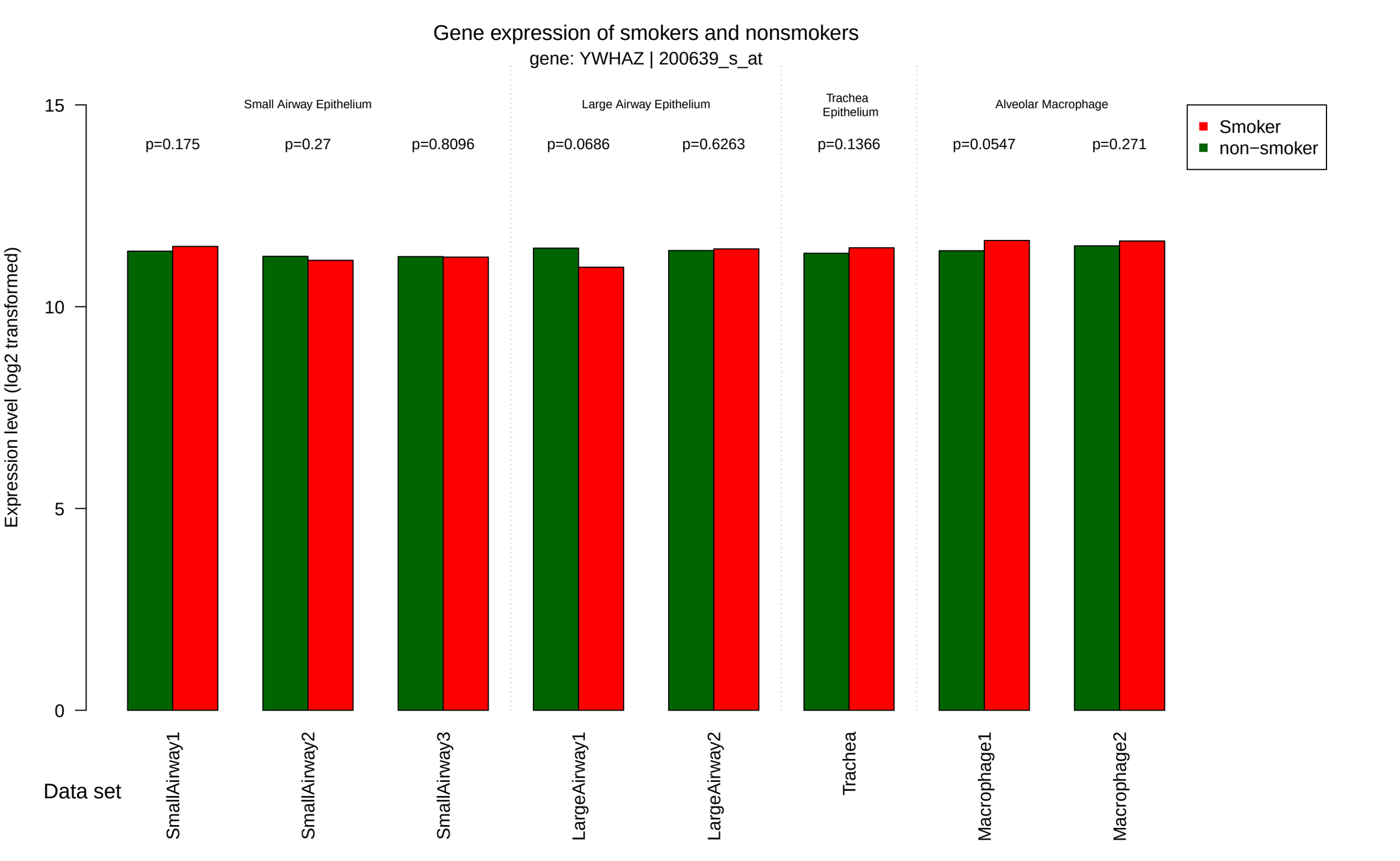
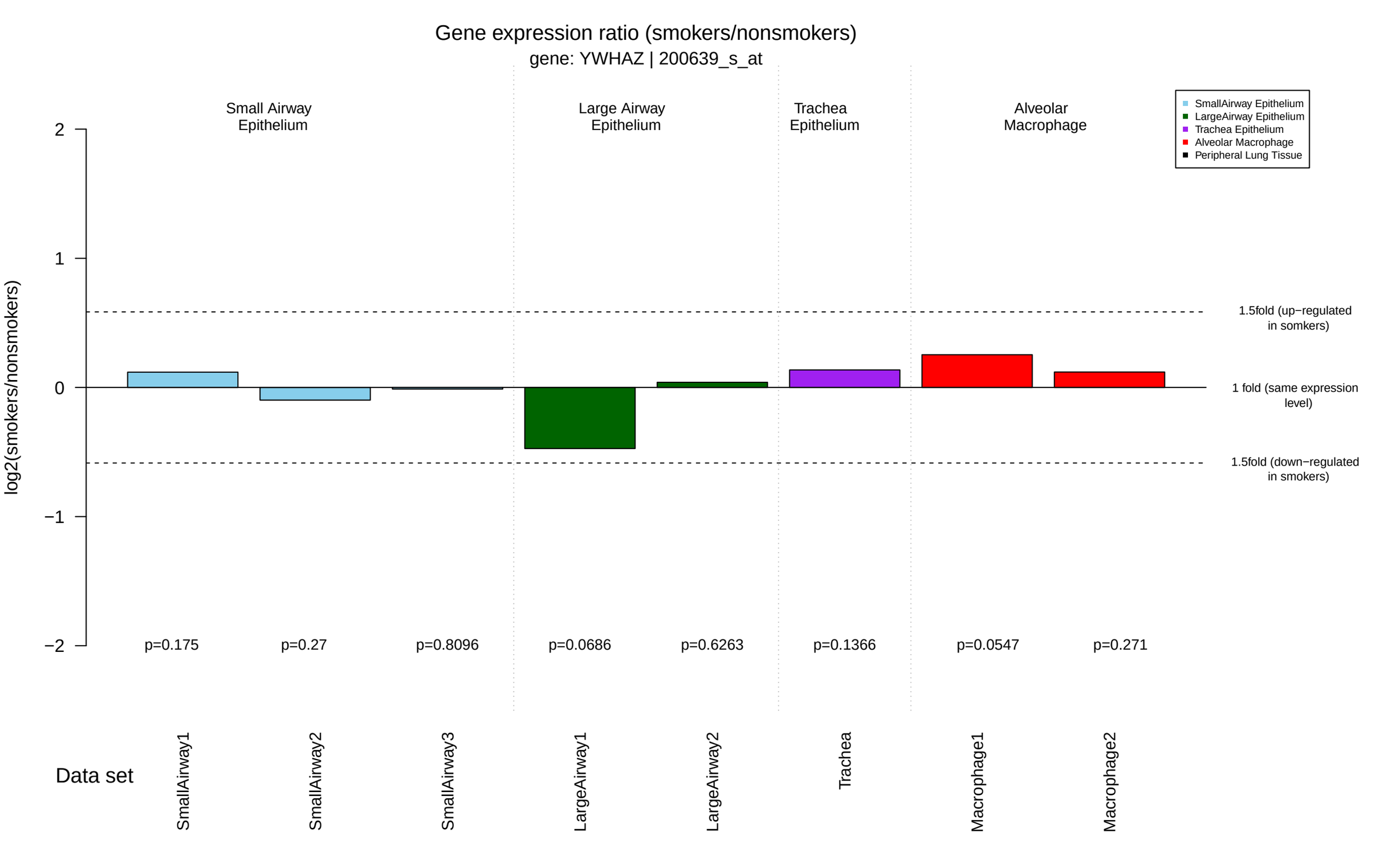
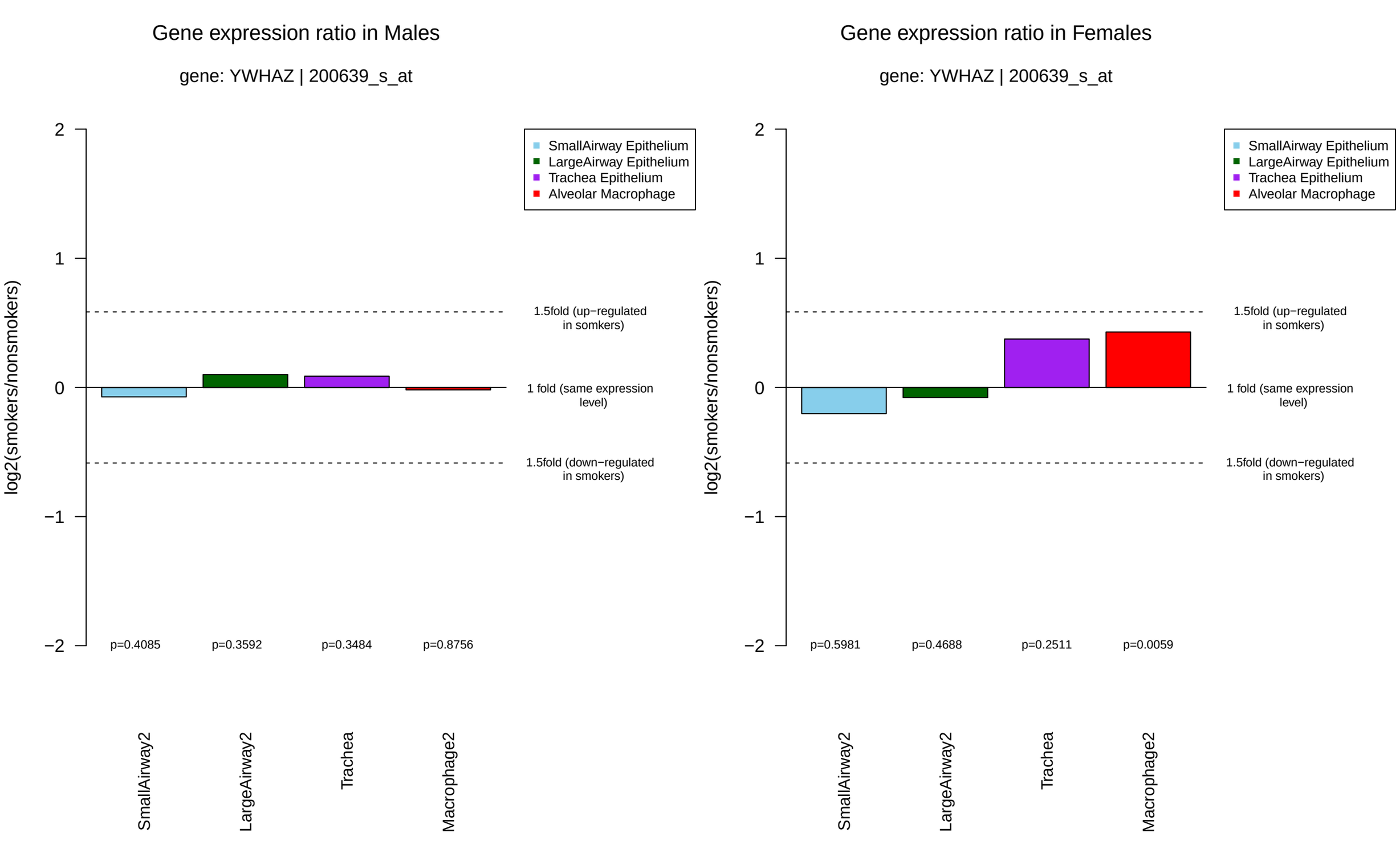
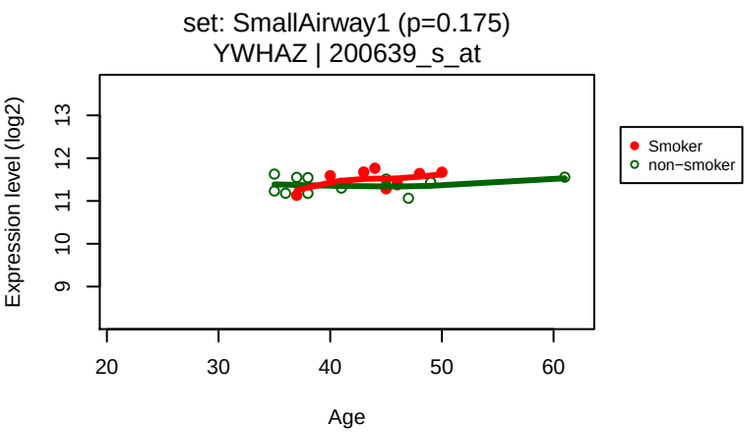
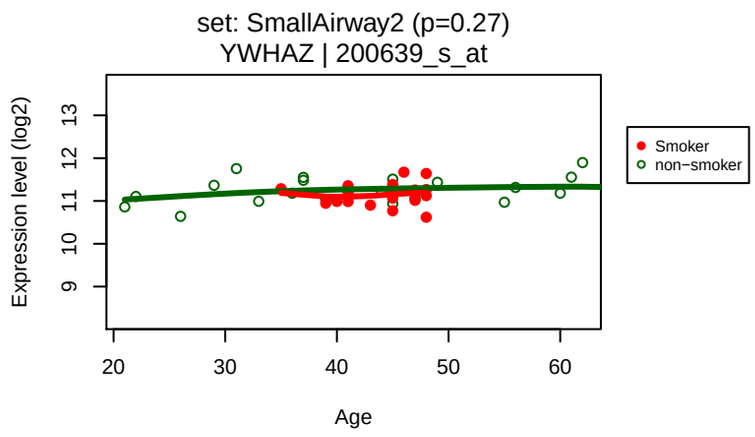
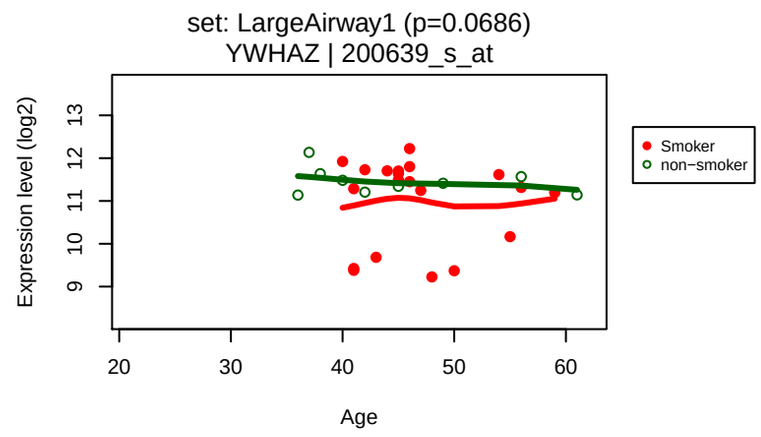

Gene expression of smokers and nonsmokers
gene: YWHAZ | 200639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.175
p=0.27
p=0.8096
p=0.0686
p=0.6263
p=0.1366
p=0.0547
p=0.271
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

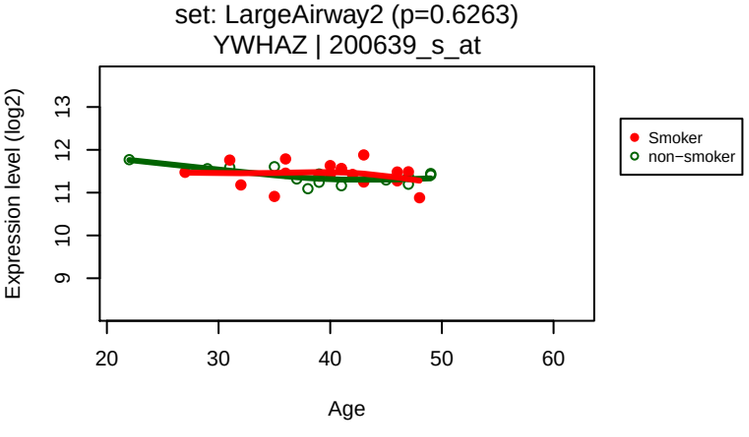

Gene expression ratio (smokers/nonsmokers)
gene: YWHAZ | 200639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.175
p=0.27
p=0.8096
p=0.0686
p=0.6263
p=0.1366
p=0.0547
p=0.271
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

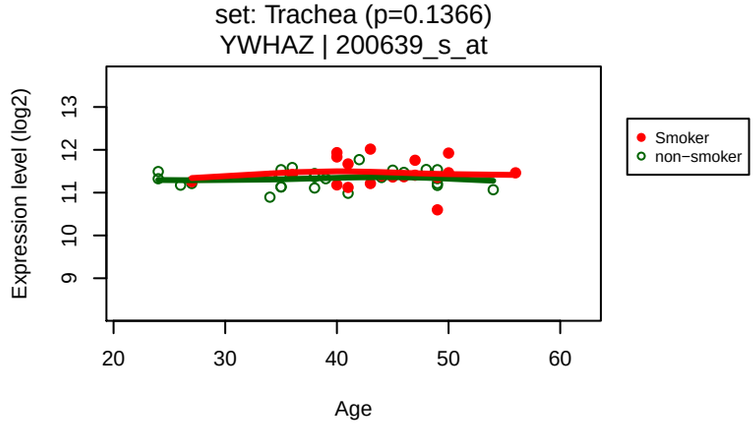

Gene expression ratio in Males
gene: YWHAZ | 200639_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4085
p=0.3592
p=0.3484
p=0.8756
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: YWHAZ | 200639_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5981
p=0.4688
p=0.2511
p=0.0059
SmallAirway2
LargeAirway2
Trachea
Macrophage2

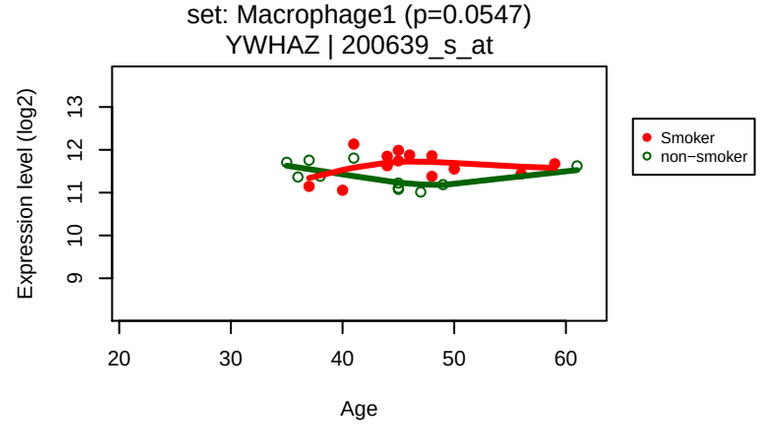

set: SmallAirway1 (p=0.175)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker

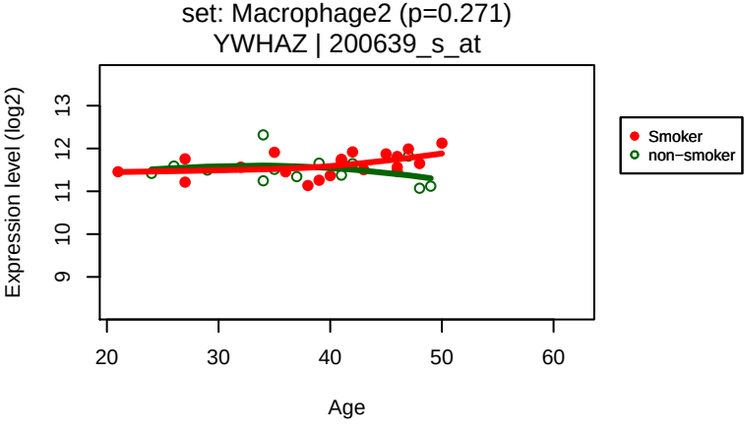

set: SmallAirway2 (p=0.27)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker

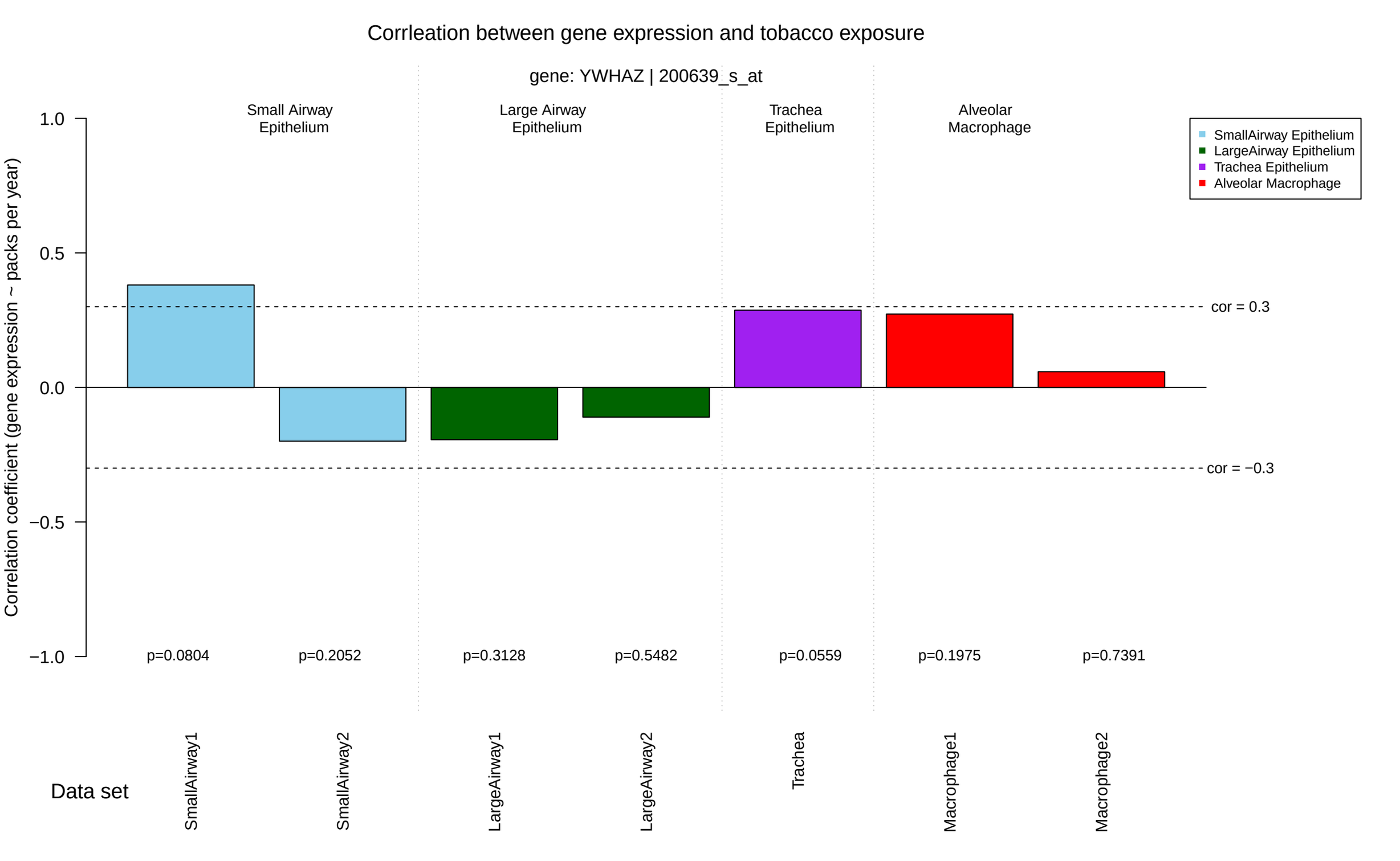

set: LargeAirway1 (p=0.0686)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6263)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1366)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0547)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.271)
YWHAZ | 200639_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: YWHAZ | 200639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0804
p=0.2052
p=0.3128
p=0.5482
p=0.0559
p=0.1975
p=0.7391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2